§3 The three principles: SM, EU and FR — part 4 of 5

Reference: subsec:sd_necessity

Source: paper/section3_principles.tex, lines 3119–4168 (§3.10 response-diversity degeneration and long-term feasibility: conditional necessity), at commit e6e12cd. The Japanese manuscript is authoritative. Part 4 of §3 follows 03_principles_3.md. Part 5 of §3 follows in 03_principles_5.md.

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§3.10 Response-diversity degeneration and long-term feasibility: conditional necessity

Reference: subsec:sd_necessity

Purpose and scope

This subsection asks how a persistent degeneration of response diversity relates to long-term feasibility. The quantity used is the response diversity $\mu_{\mathrm{eff},t}$ (def:mu_eff, classification C), a diagnostic quantity not used in the definition of any time-indexed satisfaction predicate.

The degeneration mode treated is $e_t \in E_{\mathrm{req}}(x_t)$, $A_t \neq \emptyset$, $\mu_{\mathrm{eff},t} = 1$: a feasible action exists, but the feasibility-preservation pattern over $E_{\mathrm{req}}(x_t)$ has degenerated to a single equivalence class. By the argument of prop:viab_stage_fr, $(x_t, e_t) \in \mathcal{D}_{\mathrm{FR}}$, and for $t \in \mathbb{T}_{\mathrm{P},i}$ the time-indexed predicate FR_t is true. The degeneration can therefore occur at times at which local FR is true.

What is shown is not an unconditional proposition from $\mu_{\mathrm{eff},t} = 1$ to blockage. Two steps are shown:

1. **Main proposition (no strong assumption).** If the realized external environment state satisfies the required factors, and, at feasible states, degeneration and a weak point of a feasible response persist, then inability to respond robustly persists and the trajectory stays outside the viability kernel (prop:sd_robust_nonresponse).
2. **Corollary (under the strong assumption).** Adding the strong assumption that persistence outside the kernel leads to blockage, blockage occurs at a finite time (prop:sd_necessity).

Both are statements of the direction of conditional necessity. Collapse is not derived from blockage; that relation belongs to §7. The upstream definitions of $E_{\mathrm{req}}(x)$, φ_u, $\sim$ and μ_{eff} are fixed and unchanged; $\sim_t$ denotes the relation at time t. The subsection introduces no environment subset chosen by an external analyst (rem:env_basis_intrinsic); quantification over the environment is over E_{ext} or $E_{\mathrm{req}}(x)$.

Assumption and the description of response inability

Assumption (ass:nonstationary): non-stationary environment. The realized sequence $(e_t)_{t\geq 0}$ is not constant in time: there are $t \neq t'$ with $e_t \neq e_{t'}$ ($e_t, e_{t'} \in E_{\mathrm{ext}}$). *Introduction status: O (observational). Strength: weak.* The assumption does not posit stochastic dispersion or statistical variation; it says only that the sequence is not constant. The change may come from the exogenous background component or from the action-response component (def:env_realization). It is used as a background condition for interpreting why preserving response diversity matters. The paper does not derive the existence of a weak point of a feasible response from it, and it is not used in the formal proofs of prop:sd_robust_nonresponse and prop:sd_necessity.

Definition (def:nonresponse_env_set): response-inability environment set and weak point of a feasible response (classification C). For $x \in X$, $u \in U$: $N(x,u) := \{e \in E_{\mathrm{req}}(x) \mid f(x,u,e) \notin K\} \subseteq E_{\mathrm{req}}(x)$. $N(x,u) \neq \emptyset$ says that, taking u at x, some external environment state satisfying all required factors puts the next state outside K. If at time t there is $u^\star \in A_t$ with $N(x_t, u^\star) \neq \emptyset$, then a *weak point of a feasible response* exists at t.

Relational character (`rem:nonresponse_relational`). $N(x,u)$ is determined by the triple of state, action and external environment state; it is not a property of the environment alone. The paper describes a situation in which survival fails not as attributable to the environment only, but as a relation: the state cannot respond to that environment state. A weak point is neither a constitutive requirement nor an assumption imposed at all times; it is used only as a premise of the propositions of `subsubsec:sd_main`. A state without a weak point exists (for example $E_{\mathrm{req}}(x_t) = \{e_t\}$), and the paper does not exclude it. The witness lies in $E_{\mathrm{req}}(x_t)$, not in E_{ext}, consistent with the kernel being defined by quantification over $E_{\mathrm{req}}(x)$.

Environmental difference implied by a weak point (`rem:weakness_env_difference`). With witnesses $u^\star \in A_t$, $e^\star \in N(x_t, u^\star)$: $f(x_t, u^\star, e_t) \in K$ and $f(x_t, u^\star, e^\star) \notin K$, hence $e^\star \neq e_t$ and $A(x_t, e_t) \neq A(x_t, e^\star)$. Where a weak point exists, $A(x,e)$ is non-degenerate with respect to differences of the external environment state at least at one place. The paper does not derive from this the structural environment coupling condition (`def:env_coupling`) or the choice of a particular response-coordinate set I_E.

Definition (`def:nonresponse_region`): response-inability region (classification C). $\mathcal{N}^{(1)} := \{(x,e) \in X \times E_{\mathrm{ext}} \mid A(x,e) = \emptyset\}$. $(x,e) \in \mathcal{N}^{(1)}$ is equivalent to $e \notin \mathcal{E}^{(1)}_{\mathrm{viab}}(x)$ (`def:one_step_viable_environment`); the definition lists the complement of that correspondence as pairs. Blockage $A_t = \emptyset$ is equivalent to $(x_t, e_t) \in \mathcal{N}^{(1)}$.

Definition (`def:robust_response`): robust response set and inability to respond robustly (classification C). $U^{\mathrm{rob}}(x) := \{u \in U \mid N(x,u) = \emptyset\}$. If $U^{\mathrm{rob}}(x) = \emptyset$, the state x is *unable to respond robustly*: whatever action is taken, some external environment state satisfying all required factors puts the next state outside K.

Proposition (`prop:robust_nonresponse_viab`): inability to respond robustly and the robust feasible sets. If $U^{\mathrm{rob}}(x) = \emptyset$, then $x \notin \mathrm{Viab}_f(K)$ and $x \notin \mathrm{Viab}^{(\nu)}_f(K)$ for all $\nu \geq 1$. *Proof idea:* contrapositive. Membership in the kernel (`prop:viab_core_max`) or in a finite-step set (`def:viab_stage`) yields an action with $f(x,u,e) \in K$ for all $e \in E_{\mathrm{req}}(x)$, that is $u \in U^{\mathrm{rob}}(x)$.

Non-claim on the converse. This paper does not state the converse of `prop:robust_nonresponse_viab` as a general consequence. For $E_{\mathrm{req}}(x) \neq \emptyset$, $U^{\mathrm{rob}}(x) \neq \emptyset$ expresses only that some action puts the next state in K under every required-satisfying environment state; it does not express staying in K_{req} or in the kernel. For $E_{\mathrm{req}}(x) = \emptyset$, $U^{\mathrm{rob}}(x) = U$ by definition and the non-emptiness says nothing about robustness. Inability to respond robustly ($U^{\mathrm{rob}} = \emptyset$) is also distinct from blockage ($A_t = \emptyset$): the former quantifies over $E_{\mathrm{req}}(x)$ and the latter concerns the realized e_t only (see the case analysis of `prop:sd_robust_nonresponse`).

Persistence outside K and its connection to blockage

Definition (`def:theta_exterior`): persistence outside K. Complements are taken in X: $K^c := X \setminus K$, $(\mathrm{Viab}_f(K))^c := X \setminus \mathrm{Viab}_f(K)$. The ACS *persists outside* K from t if $x_s \in K^c$ for all $s \geq t$. This excludes an overshoot, that is, a temporary exit followed by re-entry into K. The paper does not state, as a general consequence of $x_{t+T} \notin K$ at some finite time, that $x_{t+T+s} \notin K$ for all $s \geq 0$; return after leaving K is not excluded in Phase 1. A shorthand for persistence over a finite time width τ is not adopted, since without a lower bound on τ it cannot be distinguished from a temporary exit.

Assumption (`ass:theta_exterior_blockage`): persistence outside the viability kernel and its finite-time connection to blockage. For every $t \in \mathbb{N}_0$: if $x_s \notin \mathrm{Viab}_f(K)$ for all $s \geq t$, then there is $T' \in \mathbb{N}_0$ with $A_{t+T'} = \emptyset$. *Introduction status: S (structural). Strength: strong (a strong assumption).*

- The antecedent is *persistence* outside the kernel, not a temporary exit. The consequent is blockage, not collapse; the paper does not state $A_t = \emptyset \Rightarrow \mathrm{Col}_{i,t} = 1$, and the formal connection of blockage and collapse is the responsibility of §7.
- Since $\mathrm{Viab}_f(K) \subseteq K$, $K^c \subseteq (\mathrm{Viab}_f(K))^c$, so persistence outside K implies the antecedent. Placing the antecedent on the kernel rather than on K includes trajectories that re-enter K now and then without recovering robust feasibility.
- Formalizing sufficient conditions that derive this assumption from the internal dynamics

of the ACS is a Phase 2 task; no concrete path is imposed as a formal assumption here (within present framework).
- Role: used only in `prop:exterior_blockage` and in the corollaries `prop:sd_necessity` and `prop:sd_necessity_fr`; the main proposition does not use it. Trajectories that stay outside the kernel without blockage are not excluded at the level of types (`rem:exterior_without_blockage`), and the assumption removes them from the conditional analysis. Under the decomposition of §2, trajectories that keep feasible actions while staying outside the kernel, through preserved action-response components, are also removed by it.

Proposition (`prop:exterior_blockage`): persistence outside K and blockage. Under `ass:theta_exterior_blockage`, persistence outside K from t implies $A_{t+T'} = \emptyset$ for some $T' \in \mathbb{N}_0$. *Proof idea:* $K^c \subseteq (\mathrm{Viab}_f(K))^c$ and the assumption.

Persistence outside the kernel without blockage (`rem:exterior_without_blockage`). Without `ass:theta_exterior_blockage`, the paper does not state blockage as a general consequence of persistence outside the kernel. In the example of `prop:robust_noninclusion`(ii), if the exogenous background component equals b_1 at every time and $o_0 = \text{good}$, then $x_t = 1 \notin \mathrm{Viab}_f(K)$ at all times, while $A_t = \{u_\emptyset\} \neq \emptyset$ at all times. The example is constructed at the level of types and has the same reach as `rem:robust_noninclusion_reading`.

Conditional deterministic proof for response-diversity degeneration

Lemma (`lem:sd_viab`): response-diversity degeneration plus a weak point gives inability to respond robustly. Under `def:mu_eff`, `def:viab_core`, `def:nonresponse_env_set` and `def:robust_response`, if at time t: $x_t \in K$, $e_t \in E_{\mathrm{req}}(x_t)$, $A_t \neq \emptyset$, $\mu_{\mathrm{eff},t} = 1$, and a weak point of a feasible response exists at t, then $U^{\mathrm{rob}}(x_t) = \emptyset$ and $x_t \notin \mathrm{Viab}_f(K)$. *Proof idea:* (1) A witness $u^\star \in A_t$, $e^\star \in N(x_t, u^\star)$ gives $\varphi_{u^\star}(e^\star) = 0$. Since $\mu_{\mathrm{eff},t} = 1$, all of A_t has the same φ, so every $u \in A_t$ fails at $e^\star$. (2) Every $u \notin A_t$ fails at $e_t \in E_{\mathrm{req}}(x_t)$ by definition of A_t. (3) Hence for every $u \in U$ some $e \in E_{\mathrm{req}}(x_t)$ has $f(x_t, u, e) \notin K$, that is $U^{\mathrm{rob}}(x_t) = \emptyset$; by `prop:robust_nonresponse_viab` (or directly by `prop:viab_core_max`), $x_t \notin \mathrm{Viab}_f(K)$.

Logical structure of the lemma (`rem:sd_viab_structure`). The premises have different roles:

- The weak point supplies one $u^\star \in A_t$ with $N(x_t, u^\star) \neq \emptyset$.
- $\mu_{\mathrm{eff},t} = 1$ propagates that weak point to all of A_t. It is here a premise used in the derivation, not a label of a state class. Without it the conclusion fails: a type-level example with $X = \{0,1\}$, $K = \{1\}$, $U = \{u_1, u_2\}$, $E_{\mathrm{ext}} = \{e^1, e^2\}$ gives $A_t = \{u_1, u_2\}$, a weak point of u_1 at e^2, $\mu_{\mathrm{eff},t} = 2$, $u_2 \in U^{\mathrm{rob}}(1)$ and $1 \in \mathrm{Viab}_f(K)$.
- $e_t \in E_{\mathrm{req}}(x_t)$ excludes actions outside A_t by a witness inside $E_{\mathrm{req}}(x_t)$. Without it a robust action can exist outside A_t, which $\mu_{\mathrm{eff},t}$ (a cardinality over A_t) cannot assess; a second type-level example (with $e_t = e^1 \notin E_{\mathrm{req}}(1)$) shows this.
- If one assumes the stronger condition $U^{\mathrm{rob}}(x_t) = \emptyset$ directly, the same non-membership follows from `prop:robust_nonresponse_viab` without $\mu_{\mathrm{eff},t} = 1$. The lemma derives inability to respond robustly from a weak point and $\mu_{\mathrm{eff},t} = 1$.

Tendency reading of $\mu_{\mathrm{eff},t} = 1$ (`rem:mu_eff_tendency`). $\mu_{\mathrm{eff},t} = 1$ says the response pattern over $E_{\mathrm{req}}(x_t)$ is unified on A_t. The paper does not state, as a general consequence of $\mu_{\mathrm{eff},t} = 1$ alone, structural vulnerability. If the common pattern is $\varphi \equiv 1$ on $E_{\mathrm{req}}(x_t)$, the single pattern keeps feasibility one step ahead for all of $E_{\mathrm{req}}(x_t)$; this one-step property follows the temporal reach of `rem:sd_local_scope`, and the paper does not state from it long-term persistence with the same pattern at later times. (When a weak point exists, $\varphi \equiv 1$ does not occur; the case describes a boundary case without a weak point.) When a weak point exists, the common pattern contains some $e^\star$ with $\varphi(e^\star) = 0$, and $\mu_{\mathrm{eff},t} = 1$ means there is no alternative feasible action in A_t that avoids it. The non-stationarity of `ass:nonstationary` gives background for reading the importance of response diversity, but the paper does not state from non-stationarity alone that a state $e^\star$ with a weak point actually appears on the realized sequence. The case $E_{\mathrm{req}}(x_t) = \emptyset$ is outside this remark (`rem:mu_eff_empty_req`). The paper does not state blockage or collapse from $\mu_{\mathrm{eff},t} = 1$ alone; the remark is a tendency reading of vulnerability when combined with a weak point.

Internal-state differences and temporal reach (`rem:sd_internal_state_scope`). $\mu_{\mathrm{eff},t}$ is a local quantity at a fixed current state x_t. Differences of internal state that arise over time connect to the temporal discussion here only through the re-evaluation of $E_{\mathrm{req}}(x_s)$, A_s, $\mu_{\mathrm{eff},s}$ as x_s changes. The subsection does not formalize individual mechanisms of change such as aging, wear or depletion of internal resources; inter-time updating is the responsibility of §5.

Lemma (`lem:finite_exit`): loss of robust feasibility outside the viability kernel. If $x_t \in K_{\mathrm{req}}$ and $x_t \notin \mathrm{Viab}_f(K)$, then for every state-feedback policy $\pi : K_{\mathrm{req}} \to U$ there are $T \geq 1$ and a finite sequence $(e_s)_{s=t}^{t+T-1}$ with $e_s \in E_{\mathrm{req}}(x_s)$, $x_{s+1} = f(x_s, \pi(x_s), e_s)$, such that $x_{t+T} \notin K_{\mathrm{req}}$. *Proof idea:* by contradiction, the set R of states reachable from x_t under π would be an E_{req}-invariant subset of K_{req}, hence contained in $\mathrm{Viab}_f(K)$ by `prop:viab_core_max`, contradicting $x_t \notin \mathrm{Viab}_f(K)$. The sequence is a finite prefix up to the exit time; π is not evaluated at $x_{t+T} \notin K_{\mathrm{req}}$.

Reach of the lemma (`rem:uniform_bound`). The lemma asserts only the *existence* of an exit-causing finite sequence and time for each state-feedback policy. It does not assert that a realized trajectory leaves K_{req}: it quantifies over all elements of $E_{\mathrm{req}}(x_s)$, not over realizations following the exogenous background and g_O of `def:env_realization`, so it does not assert finite-time exit of a realized trajectory. It does not assert generalization to history-dependent policies, nor a uniform bound $\sup T < \infty$ over initial states and policies; these need additional structural conditions and are a Phase 2 task. The lemma is not used in the formal proofs of `prop:sd_robust_nonresponse` and `prop:sd_necessity` (within present framework).

Main proposition and corollaries: persistence of inability to respond robustly and connection to blockage

Reference: `subsubsec:sd_main`

Proposition (`prop:sd_robust_nonresponse`): main proposition — persistence of inability to respond robustly from degeneration and persistent weak points. Suppose that for each $s \geq t$: (a) $x_s \in K \Rightarrow e_s \in E_{\mathrm{req}}(x_s)$; (b) $x_s \in K$ and $A_s \neq \emptyset \Rightarrow \mu_{\mathrm{eff},s} = 1$ and a weak point of a feasible response exists at s. Then for each $s \geq t$, $x_s \notin \mathrm{Viab}_f(K)$, and at every s with $x_s \in K$, $U^{\mathrm{rob}}(x_s) = \emptyset$. The proposition does not use `ass:theta_exterior_blockage`. *Proof idea:* three cases per s. (1) $x_s \notin K$: $\mathrm{Viab}_f(K) \subseteq K$. (2) $x_s \in K$, $A_s \neq \emptyset$: `lem:sd_viab`. (3) $x_s \in K$, $A_s = \emptyset$: by (a), $e_s \in E_{\mathrm{req}}(x_s)$, and every u has $f(x_s, u, e_s) \notin K$, so $e_s \in N(x_s, u)$ for all u, giving $U^{\mathrm{rob}}(x_s) = \emptyset$ and, by `prop:robust_nonresponse_viab`, $x_s \notin \mathrm{Viab}_f(K)$.

Corollary (`prop:sd_necessity`): finite-time blockage from persistent degeneration, under the strong assumption. Under `ass:theta_exterior_blockage`, if conditions (a)(b) of `prop:sd_robust_nonresponse` hold for each $s \geq t$, then there is $T \in \mathbb{N}_0$ with $A_{t+T} = \emptyset$. *Proof idea:* the main proposition gives $x_s \notin \mathrm{Viab}_f(K)$ for all $s \geq t$, which is exactly the antecedent of the assumption.

Division of roles (`rem:sd_role_division`).

1. The main proposition follows from the definitions, `lem:sd_viab` and `prop:robust_nonresponse_viab` only, with no strong assumption. Its conclusion is persistence outside the kernel and, at feasible times, inability to respond robustly. Blockage is not in the conclusion.
2. The corollary connects this conclusion to blockage through the strong assumption; the finite-time blockage depends on that assumption.
3. Without the strong assumption, the claim of the paper is the main proposition's conclusion only. Blockage does not follow in general even when that conclusion holds (`rem:exterior_without_blockage`).

Corollary (`prop:sd_necessity_fr`): persistent degeneration and local FR. Under `ass:theta_exterior_blockage` and (a)(b) of `prop:sd_robust_nonresponse`, there is $T \in \mathbb{N}_0$ with $A_{t+T} = \emptyset$, and one of the following holds:

- (i) $x_{t+T} \notin K$, and the blockage at $t+T$ is (B1) or (B2) (`prop:blockage_dichotomy`). If $t+T \in \mathbb{T}_{\mathrm{P},i}$, then SM_{t+T} is false and either EU_{t+T} is false, or EU_{t+T} is true and FR_{t+T} is false.

- (ii) $x_{t+T} \in K$, and the blockage at $t+T$ is (B2). If $t+T \in \mathbb{T}_{\mathrm{P},i}$, then EU_{t+T} is true and FR_{t+T} is false.

Proof idea: T from `prop:sd_necessity`. For $x_{t+T} \notin K$ use `prop:blockage_dichotomy` and `def:SM`. For $x_{t+T} \in K$, (a) gives $e_{t+T} \in E_{\mathrm{req}}(x_{t+T})$, and as in the proof of `prop:viab_stage_fr`, $(x_{t+T}, e_{t+T}) \in \mathcal{D}_{\mathrm{FR}}$, so the blockage is (B2). Under (a), at a time $s \geq t$ with $x_s \in K$ no (B1) blockage occurs, and if that time lies in $\mathbb{T}_{\mathrm{P},i}$, EU_s is true. In case (i), the blockage means one-step return to K is impossible; it says nothing about possible recovery to K over several steps (`rem:eu_one_step_scope`).

Role and strength of condition (a) (`rem:req_condition_role`). Condition (a) requires that at feasible times the realized external environment state satisfies the required factors. It is needed to apply `lem:sd_viab` at each time. The paper does not derive it from CR-1: (CR-1b) requires only that some x have $E_{\mathrm{req}}(x) \neq \emptyset$, not that the realized state lie in $E_{\mathrm{req}}(x_s)$ at each time. Condition (a) is therefore a limiting condition in the antecedent and is not treated as generally true. For trajectories containing times at which (a) fails, `prop:sd_robust_nonresponse`, `prop:sd_necessity` and `prop:sd_necessity_fr` say nothing.

Degeneration mode treated (`rem:sd_failure_mode_scope`). If $A_t = \emptyset$ then $\mu_{\mathrm{eff},t} = 0$ (`prop:fr_mu_eff`) and this is already blockage. If $E_{\mathrm{req}}(x_t) = \emptyset$ and $A_t \neq \emptyset$ then $\mu_{\mathrm{eff},t} = 1$, but this is a set-theoretic consequence of there being no required-satisfying states to compare (`rem:mu_eff_empty_req`). The lemma, main proposition and corollary treat the degeneration mode with $e_t \in E_{\mathrm{req}}(x_t)$, $A_t \neq \emptyset$, $\mu_{\mathrm{eff},t} = 1$. The conclusion is not extended to all cases with $\mu_{\mathrm{eff},t} \leq 1$. Falsity of local FR is blockage itself on $\mathcal{D}_{\mathrm{FR}}$ (`def:FR`); the propositions derive nothing from falsity of local FR.

Responses that stay in the kernel, and the difference in outcome

The main proposition describes persistence of states that cannot respond. As its counterpart, the following states, without assumptions, what follows when responses that stay in or enter the kernel can be chosen.

Definition (`def:kernel_response`): kernel-maintaining response set (classification C). $U^{\mathrm{Viab}}(x) := \{u \in U \mid \forall e \in E_{\mathrm{req}}(x) : f(x,u,e) \in \mathrm{Viab}_f(K)\}$. For $x \in \mathrm{Viab}_f(K)$ its elements are responses that stay in the kernel; for $x \notin \mathrm{Viab}_f(K)$ they are responses that enter the kernel under every required-satisfying environment state. If $x \in \mathrm{Viab}_f(K)$ then $U^{\mathrm{Viab}}(x) \neq \emptyset$ (`prop:viab_core_max`); since $\mathrm{Viab}_f(K) \subseteq K$, $U^{\mathrm{Viab}}(x) \subseteq U^{\mathrm{rob}}(x)$; if $E_{\mathrm{req}}(x) = \emptyset$ then $U^{\mathrm{Viab}}(x) = U$.

Proposition (`prop:kernel_maintenance`): trajectories that keep choosing kernel-maintaining responses. If $x_T \in \mathrm{Viab}_f(K)$ and, for all $s \geq T$, $e_s \in E_{\mathrm{req}}(x_s)$ and $u_s \in U^{\mathrm{Viab}}(x_s)$, then for all $s \geq T$, $x_s \in \mathrm{Viab}_f(K)$ and $A_s \neq \emptyset$. No assumption is used. *Proof idea:* induction on s; $x_{s+1} = f(x_s, u_s, e_s) \in \mathrm{Viab}_f(K) \subseteq K$, hence $u_s \in A_s$.

Definition (`def:kernel_response_acquisition`): acquisition of kernel-maintaining responses (classification C). A trajectory *has acquired kernel-maintaining responses at time* T if for all $s \geq T$, $E_{\mathrm{req}}(x_s) \neq \emptyset$ and $u_s \in U^{\mathrm{Viab}}(x_s)$. The condition $E_{\mathrm{req}}(x_s) \neq \emptyset$ is included because for $E_{\mathrm{req}}(x_s) = \emptyset$, $U^{\mathrm{Viab}}(x_s) = U$ and the membership would hold vacuously. This is a property of the trajectory; it may or may not hold through the generation of the effective action u_s (§4) and the update of the internal model (§5).

Difference in outcome and non-claims (`rem:kernel_response_scope`). `prop:kernel_maintenance` and `prop:sd_robust_nonresponse` separate two cases in one framework. If a kernel-maintaining response exists and is chosen continually, no blockage occurs. If inability to respond robustly persists at feasible states, no kernel-maintaining response exists. What happens afterwards in the latter case depends on the sequence of realized external environment states and on the actions chosen, and the connection to blockage is stated only as a corollary under the strong assumption. This paper does not state any of the following as a general consequence:

1. that acquisition of kernel-maintaining responses occurs, or an upper bound on its time;
2. that updating the internal model (learning) leads to acquisition of kernel-maintaining

responses;
3. that $U^{\mathrm{Viab}}(x) \neq \emptyset$ implies the action actually chosen lies in $U^{\mathrm{Viab}}(x)$.

The ACS is not required to hold or compute $\mathrm{Viab}_f(K)$ or $U^{\mathrm{Viab}}(x)$ internally (`subsec:description_levels_global`).

Logical status of the results

Meaning of the necessity direction (`rem:sd_necessity_scope`). `prop:sd_necessity` does not assert that blockage is reached merely because $\mu_{\mathrm{eff},t} = 1$ at one time. What it shows is the direction of a conditional necessity: when (a), (b) (including a weak point) and `ass:theta_exterior_blockage` hold, the degeneration $\mu_{\mathrm{eff},s} = 1$ cannot persist indefinitely without reaching blockage. The result of Phase 1 is therefore not a proof that $\mu_{\mathrm{eff},t} \geq 2$ at each time is an unconditional universal necessary condition. What is shown is the structural consequence that, under the stated additional conditions, avoiding persistent degeneration of response diversity is conditionally required for long-term non-blockage; this describes part of the long-term aspect of FR under an additional assumption. Without the strong assumption, only the main proposition's conclusion (inability to respond robustly and persistence outside the kernel) can be shown. For times with $E_{\mathrm{req}}(x_t) = \emptyset$ the proposition does not conclude blockage, and the paper does not state $A_t = \emptyset$ or collapse from that fact. If any of the additional conditions fails, the proposition says nothing about blockage.

Absence of circular reference (`rem:no_circular`). The formal proof of the main proposition depends only on `def:at`, `def:mu_eff`, `def:req_domain`, `def:viab_core`, `def:nonresponse_env_set`, `def:robust_response`, `prop:viab_core_max`, `prop:robust_nonresponse_viab` and `lem:sd_viab`, and uses no assumption. The corollary `prop:sd_necessity` adds `ass:theta_exterior_blockage`; `prop:sd_necessity_fr` further adds `def:E_req`, `def:eu_fire`, `def:fr_domain`, `def:SM` and `def:FR`; `prop:kernel_maintenance` depends only on `def:at`, `def:viab_core` and `def:kernel_response`. `ass:nonstationary` and `rem:viability_relation` (a conceptual placement relative to viability theory) are not used in these proofs, and neither are `lem:finite_exit` and `prop:exterior_blockage`. The dependencies use no concept later than `subsec:sd_necessity`, so no circular dependence on later sections arises.

Logical status of the conditional necessity (`rem:sd_status`). The link between degeneration of response diversity and long-term feasibility has a logical status different from the definitional equivalence of SM (`prop:sm_K_equiv`) and from the definitional coincidence of local FR with blockage (`def:FR`). SM and local FR are time-indexed satisfaction predicates at a single time. The response diversity $\mu_{\mathrm{eff},t}$ is a classification-C diagnostic quantity expressing how many modes of one-step feasibility preservation exist over differences of required-satisfying environment states, and is used in the definition of no time-indexed predicate. The paper does not state, as a general consequence of the value of $\mu_{\mathrm{eff},t}$ at one time alone, a consequence about maintaining response diversity in the long run. What was shown conditionally and deterministically is neither a sufficient condition for long-term maintenance of response diversity nor blockage from $\mu_{\mathrm{eff},t} \leq 1$ in general. It is (main proposition) the conditional consequence of inability to respond robustly and persistence outside the kernel when states satisfying the required factors continue and, at feasible states, a weak point and degeneration persist; and (corollary) finite-time blockage when the strong connection assumption is added. `prop:sd_necessity_fr` separates whether the blockage time is one at which $x \notin K$ or one at which local FR is false. The paper does not define the three principles as an algorithm that computes the kernel and moves toward it. SM describes the feasibility of the current state, EU the satisfaction of critical required factors at each time, and local FR the existence of a feasible action in a situation in $\mathcal{D}_{\mathrm{FR}}$; the relation is only that these can be consistent with long-term feasibility maintenance. The relation to finite-step robustness is limited to one direction of `prop:viab_stage_fr`. The paper does not state "if all time-indexed satisfaction predicates are true then the state stays in the kernel", nor "if the state is in the kernel then all time-indexed satisfaction predicates are true", and does not elevate membership in the kernel to a condition for an ACS to hold. The result clarifies the range provable in Phase 1 as the direction of conditional necessity against persistent degeneration of response diversity (claim strength ≤ proof strength).

Degree propositions are a Phase 2 task (`rem:sd_degree`). `prop:sd_necessity` does not assert a monotone relation between the value of $\mu_{\mathrm{eff},t}$ and the time to blockage, for example "the larger the response diversity, the longer until the feasible action set becomes empty". Formalizing such a degree proposition needs additional dynamical assumptions, a comparison principle, time-average conditions or a stochastic formulation. What Phase 1 establishes is the conditional consequences above, not a quantitative monotone relation. The paper does not state blockage from $\mu_{\mathrm{eff},t} = 1$ alone.

Improvability of the feasibility margin is an auxiliary diagnostic (`rem:sd1_unfulfilled`). The condition $\exists u \in A_t : \Pi(f(x_t, u, e_t)) > \Pi(x_t)$ introduced in `rem:sd_improvement_aux` describes whether the feasibility margin can be improved by one step under the current realized external environment state. It is neither the condition of local FR (`def:FR`) nor an antecedent of the propositions of this subsection. The paper does not state, as a general consequence of its failure alone, falsity of local FR, $\mu_{\mathrm{eff},t} \leq 1$, blockage or future loss of feasibility: for example, at a high feasibility margin where no action raises Π further, $E_{\mathrm{req}}(x_t) \neq \emptyset$ and $\mu_{\mathrm{eff},t} \geq 2$ can hold. Conversely, the existence of an action that improves the margin does not by itself yield, as a general consequence, non-degeneracy of response diversity or long-term feasibility.

Measure-theoretic placement of μ_{eff} and its delegation to Phase 2 (`rem:mu_measure`). $\mu_{\mathrm{eff}}(x, e)$ is the cardinality of the quotient $A(x, e)/\sim$ (`def:mu_eff`). In Phase 1 it is used as a cardinality diagnostic of the number of modes of feasibility preservation; finiteness is not assumed and no threshold on μ_{eff} is placed in any time-indexed predicate. For continuous $A(x, e)$, quantifying diversity as a continuous quantity would need additional structure such as measurability of equivalence classes or of the quotient space. Refinement of μ_{eff} by a measure, a quotient measure, a covering number, an effective dimension or another continuous diversity index is a Phase 2 task (within present framework).

Check status. This summary adds no new definition, assumption or result. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.